Strategic Splitting under Hamilton's Apportionment Method

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Abstract

We study the strategic manipulation of Hamilton's apportionment method through population splitting, modelling the process as a sequential game in which states partition their populations to maximise allocated seats. We prove that the unique Nash Equilibrium is identical to Jefferson's method: every state receives $\lfloor q_i/r \rfloor$ seats, where r is the equilibrium threshold. The optimal strategy is efficiently computable via binary search in $O(n \log(1/\varepsilon))$ time. Monte Carlo experiments across Uniform, Dirichlet, and Power Law population distributions reveal near-universal manipulation opportunities under skewed distributions, with the Dirichlet regime providing an informative intermediate case between the two extremes. Applied to Indian Lok Sabha data, seven large states gain seats at the expense of eight smaller ones, though the practical cost of splitting makes systematic exploitation politically implausible.

1 Introduction

Apportionment is necessary in a democratic nation to ensure fair representation of the population. Fairness is achieved when the number of people represented by a seat falls within a specific quota for the region. This process is crucial for maintaining both political balance and legitimacy.

Hamilton's method of apportionment first assigns seats based on a state's quota, which is its population-based entitlement. Initially, the "floor" (the integer part of the quota) is assigned. The remaining seats are then allocated based on the largest fractional remainders.

This inherent rounding in the Hamilton method creates an opportunity for strategic manipulation, known as "Splitting." In this manipulation, a single state may divide into multiple smaller states before the seat allocation process. Each resulting fragment is then independently assigned its own seat quota and remainder. The total number of seats acquired by the fragmented parts can, as a result, exceed the number of seats the original, unified state would have been allocated, thus maximizing the overall seat count.

A similar strategy could be applied to electoral voting. A party could estimate the election outcome using past vote data or campaign information, and then strategically choose to split the candidate list into multiple sub-lists to gain extra seats through this division. Consequently, the total number of seats gained through all these sub-lists is greater than what would have been gained with the original single list. This phenomenon is often referred to as Superadditivity.

The 2012 Hong Kong Legislative Council elections offer a compelling, real-world example of a flaw in electoral systems. The election utilized the Hare quota with the largest remainder (HQLR) method, mathematically very similar to Hamilton's method, to allocate seats based on votes.

A key incident involved the pro-Beijing Democratic Alliance for the Betterment and Progress of Hong Kong (DAB), which employed a strategy of "list splitting" in multiple constituencies. Specifically, in the Hong Kong Island constituency, the DAB divided its candidates into two coordinated lists, receiving 33,901 and 36,517 votes, respectively.

This tactic successfully exploited the remainder allocation rule. Consequently, the DAB secured two seats despite receiving a lower total number of votes than the Civic Party, which won only one seat. This outcome clearly demonstrates how list splitting can manipulate the largest remainder method to gain an advantage.

Something similar can happen in the case of states. A state can fragment itself into multiple sub-states, and all of these sub-states combined may gain more representation than if they had remained a single entity. A related example can be seen in the political restructuring of Nigeria during the 1970s. During the country's transition toward a more representative political system, Nigeria was initially organized into major regional divisions, which were later subdivided into

multiple states. Because representation and federal resource allocation were tied to the number of states and administrative units, political actors had incentives to redraw boundaries and create additional states in order to increase their representation and access to resources.

To analyze the phenomenon of strategic splitting, we modeled it as a Game Theoretic problem. This framework allows states to fragment their populations strategically to maximize their total number of legislative seats. Furthermore, the states' actions are sequenced in descending order of their population. Crucially, each state possesses complete information regarding the actions of preceding states, and once a state finalizes its population configuration, that configuration is immutable.

The aim of this framework is to identify the optimal splitting strategies for all states. We hypothesize that all states will adopt a splitting strategy that collectively maximizes their total seat allocation. A key feature is that states move in descending order of population, meaning early movers must strategically anticipate the responses of smaller states. This necessitates strategic foresight and equilibrium behavior among all states.

2 Project Objectives

This project investigates strategic manipulation under Hamilton's apportionment method, where states fragment their populations into smaller units to collectively secure more seats than they would receive as a unified entity. Our objectives span theoretical analysis, algorithm design, and empirical validation.

1. **Equilibrium Analysis in the Sequential Game.** Model strategic splitting as a sequential game in which states partition their populations to maximize seat counts, and characterize the Nash equilibrium and Subgame Perfect Equilibrium of this game. Specifically, determine whether equilibrium strategies are unique, whether move order affects outcomes, and how the equilibrium seat allocation relates to classical apportionment methods.

2. **Algorithm Design and Complexity Analysis.** Develop efficient algorithms for computing optimal splitting strategies and the equilibrium divisor r^*. Analyze the computational complexity of these algorithms and establish tractability guarantees for realistic problem sizes.

3. **Empirical Validation via Simulation.** Quantify the frequency and magnitude of manipulation opportunities through Monte Carlo simulations across synthetic population distributions (Uniform, Dirichlet, Power Law) and varying parameters n and H. Measure how often Hamilton and Jefferson diverge, how many fragments manipulation requires in practice, and how these quantities scale with population inequality.

4. **Real-World Case Studies.** Validate the theoretical framework on real apportionment instances: India's 2011 Lok Sabha apportionment ($H = 543$, $n = 35$ states and UTs) and the 2012 Hong Kong Legislative Council election. Identify which states gain or lose seats under strategic splitting, and assess the political feasibility of manipulation at scale.

5. **Simultaneous-Move Variant.** Analyze how equilibrium strategies and seat allocations change when all states split simultaneously without observing each other's moves, and compare outcomes with the sequential game.

3 Previous Work

1. **Balinski and Young – The Theory of Apportionment: Balinski and Young [1980]** This work provides the fundamental theoretical framework for the study of apportionment. It formalizes the apportionment problem and introduces fairness criteria such as quota, population monotonicity, and house monotonicity. The paper also analyzes classical apportionment methods including Hamilton, Jefferson (D'Hondt), Webster (Sainte-Laguë), Adams, and Hill methods, and establishes important results regarding their properties and paradoxes.

2. **Bredereck et al. Strategic Campaigns in Apportionment Elections: Bredereck et al. [2026]** This paper studies how election outcomes can be strategically influenced by changing a limited number of votes. It models this as a bribery or strategic campaigning problem and analyzes the computational complexity of determining the minimum number of voters that must change their vote to alter the seat allocation. The work is closely related to our research as it also examines how small changes in voter distribution can affect the final apportionment outcome.

3. **Hong Kong Election Case Study: Carey [2017]** A well-documented real-world example of this occurred in the 2012 Hong Kong Legislative Council elections Carey [2017]. The electoral system used the Hare quota with the largest remainder method (HQLR), which is mathematically very similar to Hamilton's method. In this system, the quota (price of a seat) is calculated by dividing the total votes by the number of seats. Each party first receives seats equal to the number of full quotas in its vote total, and the remaining seats are allocated to parties with the largest remainders.

   This mechanism effectively means that some seats are obtained at a lower "price" through the remainder allocation, leading to differences in the number of votes required per seat across parties. As a result, the system can be strategically manipulated, and our paper present way to analyse those.

4. **Coalitional Game Theory for Communication Networks: Saad et al. [2009]**

   We refer to this paper to gain a proper understanding of how game theory works, since we were not very familiar with the definitions of concepts such as a game, players, and payoffs. This paper presents these ideas in a clear and accessible manner, discussing various directions of analysis through several examples.

   The paper explains coalitional game theory and how it can be applied in wireless communication networks. It also discusses non-cooperative game theory, where players act independently and make decisions based solely on their own interests.

   Further, the paper introduces the definition and properties of coalitions, explaining why players may choose to form coalitions and how cooperation can improve outcomes. An important point discussed is how the structure of communication between players can influence the payoff of a coalition. For example, if there are three players A, B, and C, the payoff may differ depending on how these players are connected. If there is no direct link between A and C, communication may have to occur through B, which can change the efficiency of interaction and lead to different payoff outcomes.

4 Defining the game

The players are the states in the game. Each state aims to maximize the number of seats it receives. A state may split its population into multiple fragments before seats are allocated. If a state splits into k fragments, then the total number of seats received by that state is equal to the sum of the seats allocated to its fragments.

The game proceeds sequentially as follows.States are ordered in decreasing order of population. Then the states take turn one by one starting from the state with largest population.A state has only one oppurtunity to split.

When a state moves, it chooses how to partition its population into fragments.They know the splitting of all the states which hasd turns before it.Each state chooses its partition strategically in order to maximize its total number of seats.

After all states have chosen their partitions, seats are allocated using Hamilton's method applied to all resulting fragments.
The number of seats obtained by a state is defined as the sum of the seats allocated to all fragments originating from that state.

5 Technical Preliminaries

5.1 Fragment Reduction Lemma

Lemma 1 (Fragment Reduction Lemma)**.** *Let a state split its population into fragments $S_1, S_2, \ldots, S_k$. Then there exists an equivalent representation $X_1, X_2, \ldots, X_j$ such that $0 < X_i \leq 1$ for all i, and the total number of seats obtained under Hamilton's method remains unchanged.*

Proof. Write each fragment as $S_i = \lfloor S_i \rfloor + (S_i - \lfloor S_i \rfloor)$. Replace S_i by $\lfloor S_i \rfloor$ copies of 1 together with the fractional part $S_i - \lfloor S_i \rfloor$. Applying this transformation to every fragment yields $X_1, \ldots, X_j$ with $0 < X_i \leq 1$. Since any fragment of size at most 1 can receive at most one seat under Hamilton's method, the total number of seats allocated to the state remains unchanged. □

Example. A split $(2.4, 1.3)$ can be rewritten as $(1, 1, 0.4, 1, 0.3)$.
Hence, without loss of generality, we assume a state's fragments satisfy

$$1 \geq X_1 \geq X_2 \geq \cdots \geq X_j > 0.$$

Each fragment can therefore receive either one seat or zero seats, which significantly simplifies the analysis of optimal strategies.

Lemma 2 (Monotonicity of Seat Allocation)**.** *Let the reduced fragments of a state be ordered as*

$$1 \geq X_1 \geq X_2 \geq \cdots \geq X_j > 0.$$

If fragment X_i receives a seat under Hamilton's method, then every fragment X_k with $k < i$ also receives a seat.

Proof. Seats are allocated to fragments with the largest remainders. Since all fragments have size at most 1, each fragment's remainder equals its size. We consider two cases based on the value of X_1.

Case 1: $X_1 = 1$**.** A fragment of size exactly 1 receives its seat as a guaranteed lower quota seat, independent of the remainder competition. Since $X_k \geq X_i$ for all $k < i$, every such fragment

either receives a lower quota seat (if $X_k = 1$) or has remainder $X_k \geq X_i$. In either case, X_k receives a seat whenever X_i does.

Case 2: $X_1 < 1$**.** All fragments have size strictly less than 1, so every fragment receives zero lower quota seats and competes purely on remainders. The remainder of each fragment equals its size, so the remainder of X_k satisfies $X_k \geq X_i$ for all $k < i$. Since Hamilton's method allocates remainder seats in decreasing order of remainder value, X_k receives a seat whenever X_i does.

In both cases, if X_i receives a seat then every X_k with $k < i$ also receives a seat. □

6 Insights from the Two-Player Game

We now analyze the game in the case of two states. Let the quotas of the two states be

$$q_A = A + x, \qquad q_B = B + y,$$

where $A, B \in \mathbb{N}$ and $0 \leq x, y < 1$.

6.1 Seat Bounds

Theorem 1. *In the two–player game, the number of seats s_A obtained by state A satisfies*

$$A \leq s_A \leq A + 1,$$

and similarly $B \leq s_B \leq B + 1$.

Proof. A state can always guarantee its integer seats by not splitting. Thus state A can always secure at least A seats and state B can secure at least B seats.

Since the quota of A is $A + x < A + 1$, Hamilton's method cannot allocate more than $A + 1$ seats to A. The same argument holds for B. □

Thus each player is effectively competing only for the additional remainder seat.

6.2 Winning the Extra Seat

We now characterize when a player can obtain the additional seat.

By the Fragment Reduction Lemma, we may assume the splitting strategy of A is

$$X_1 \geq X_2 \geq \cdots \geq X_k, \qquad 0 < X_i \leq 1.$$

If A wishes to obtain $A+1$ seats, it must split into at least $A+1$ fragments. Fragments beyond the first $A+1$ cannot receive seats, so it suffices to consider splitting into exactly $A+1$ fragments.

Let the smallest of these fragments be X_{A+1}. Among all ways to partition quota $A+x$ into $A+1$ parts, the minimum part is maximized when all parts are equal, by the inequality of arithmetic and geometric means. This yields

$$X_{A+1} \leq \frac{A + x}{A + 1},$$

with equality when A splits into $A + 1$ equal parts. Thus the best strategy for A to secure $A + 1$ seats is to split into $A + 1$ equal parts.

Note that if $x = 0$, state A has an integer quota and its remainder is zero. In this case splitting does not help, as each equal fragment has size $\frac{A}{A+1} < 1$ and the total remainder contribution is zero.

Now consider state B. If B attempts to obtain $B+1$ seats, it similarly splits into $B+1$ equal fragments, each of size

$$\frac{B+y}{B+1}.$$

State B obtains the extra seat if and only if its fragment size exceeds that of A, i.e.,

$$\frac{B+y}{B+1} > \frac{A+x}{A+1}.$$

6.3 Outcome of the Two-Player Game

We now summarize the optimal strategies derived from the previous analysis.

1. If $\dfrac{A+x}{A+1} > \dfrac{B+y}{B+1}$, then the unique optimal strategy is for state A to split into $A+1$ equal parts, while state B does not split.
2. If $\dfrac{B+y}{B+1} > \dfrac{A+x}{A+1}$, then the unique optimal strategy is for state B to split into $B+1$ equal parts, while state A does not split.
3. If $\dfrac{A+x}{A+1} = \dfrac{B+y}{B+1}$, then the optimal strategy will depend on the tie breaking rule.

6.4 Observations

Proposition 1. *In the two–player case, the ability of a state to obtain the additional seat depends only on the quotas $q_A = A + x$ and $q_B = B + y$, and not on which state moves first.*

Proof. From the previous section, state A secures $A+1$ seats if and only if

$$\frac{A+x}{A+1} \geq \frac{B+y}{B+1}.$$

This condition depends only on q_A and q_B. Since neither quantity changes based on which state moves first, the outcome is determined solely by the quotas regardless of move order. □

Proposition 2. *Splitting favours larger states in the two-state scenario.*

Proof. Rewriting the condition,

$$\frac{B+y}{B+1} \leq \frac{A+x}{A+1}$$

is equivalent to

$$1 - \frac{y}{B+1} \geq 1 - \frac{x}{A+1}.$$

The term $\frac{x}{A+1}$ represents the loss incurred when the quota of state A is divided into $A+1$ parts. As A becomes larger, the denominator $A+1$ increases, making the loss smaller. Consequently, larger states find it easier to obtain the additional seat through splitting. □

Example. Consider quotas $q_A = 8.2$ and $q_B = 1.8$. If state A splits into 9 equal fragments, each fragment has size $8.2/9 \approx 0.91$. State B can split into at most 2 fragments of size $1.8/2 = 0.9$. Since $0.91 > 0.9$, the fragments of state A dominate those of B, and state A obtains 9 seats while state B receives only 1 seat despite having quota 1.8. This illustrates that strategic splitting favours larger states in the two-state scenario.

7 Some Observations

Observation 1 (Upper Quota Can Be Exceeded)**.** *Strategic splitting allows a state to obtain strictly more than* $\lceil q_i \rceil$ *seats. Hamilton's method does* not *guarantee upper-quota compliance.*

Proof. Consider five states with quotas (3.9, 1.2, 1.3, 1.3, 1.3) and $H = 9$. If state 1 splits into five equal fragments each with quota $3.9/5 = 0.78$, observe that every other state has at most one fragment whose remainder can compete — namely 0.2, 0.3, 0.3, 0.3 respectively. Since $0.78 > 0.3$, all five fragments of state 1 rank above every fragment of the remaining states in the remainder ordering, regardless of how those states split. State 1 therefore obtains 5 seats in every possible outcome, while the remaining four states share the remaining 4 seats, yielding the allocation (5, 1, 1, 1, 1). This exceeds $\lceil 3.9 \rceil = 4$. □

Observation 2 (Lower Quota is Guaranteed)**.** *For any state* i*, regardless of how other states split,* $s_i \geq \lfloor q_i \rfloor$.

Proof. Choosing not to split always yields at least $\lfloor q_i \rfloor$ seats under Hamilton's method. Since every state plays optimally, no state adopts a strategy that yields fewer seats than this baseline. □

Observation 3 (Bound on the Number of Fragments)**.** *Let* $M_i = H - \sum_{j \neq i} \lfloor q_j \rfloor$*. State* i *never needs more than* M_i *fragments; splitting into more is never beneficial.*

Proof. By Observation 2, every other state j receives at least $\lfloor q_j \rfloor$ seats, leaving at most M_i seats available for state i. Since each fragment wins at most one seat (Lemma 1), any fragment beyond M_i is guaranteed to receive no seat, and therefore splitting further is always suboptimal. □

Observation 3 implies the strategy space is finite and polynomial in H, making the game tractable in principle.

8 Synthetic Simulation

To gain intuition before developing the general theory, we ran a series of small synthetic simulations. For each instance, we generated random population quotas, computed the standard Hamilton allocation, and then determined the optimal splitting strategy via exhaustive backward induction over the sequential game. The optimal allocation was recorded and compared against the Hamilton baseline.

8.1 A Worked Example

We present one representative instance in detail. Consider $n = 5$ states with total seats $H = 20$ and population quotas as given in Table 1.

State	Population	Quota	Hamilton	After Splitting
A	42	8.4	8	**9**
B	28	5.6	6	5
C	16	3.2	3	3
D	9	1.8	2	2
E	5	1.0	1	1
Total	**100**	**20.0**	**20**	**20**

Table 1: Hamilton allocation vs. optimal strategic splitting for a five-state instance with $H = 20$.

Under Hamilton's method, state B receives a remainder seat due to its fractional part of 0.6 being the second-largest remainder. However, under optimal play, state A can split into 9 equal fragments each of size $8.4/9 \approx 0.933$. By this strategy, the number of seats for A, come out to be 9 in the optimal scenerio.

A key observation we noticed in most of the simulations we ran are that the states which gained seats were the larger ones, and the states which lost seats are the smaller ones.

9 Main Theoretical Results

9.1

Definition 1 (Equilibrium threshold)**.** *Let r be the* smallest *value satisfying*

$$\sum_{i=1}^{n} \left\lfloor \frac{q_i}{r} \right\rfloor \;=\; H.$$

We call r the ***equilibrium threshold****. Equivalently, $r = D/P$ where D is Jefferson's divisor and P is the total population.*

Three key properties follow directly:

1. $r \leq 1$, since $\sum_i q_i = H$ implies the threshold cannot exceed 1.
2. r is the smallest remainder gaining a seat in equilibrium.
3. $r = D/P$ where D is Jefferson's divisor and P is total population.

9.2 Theorem 1 — Equal Splitting is Always Optimal

Theorem 2 (Equal splitting dominates)**.** *Fix all other states' strategies. Suppose some strategy $\mathcal{S}$ gives state i exactly k seats. Then equal splitting into k fragments of size q_i/k weakly dominates $\mathcal{S}$.*

Proof. Suppose some strategy $\mathcal{S}$ gives state i exactly k seats. We show that equal splitting into k fragments of size q_i/k achieves at least k seats.

Step 1. Apply Lemma 1. Order the reduced fragments as $x_1 \geq x_2 \geq \cdots \geq x_k \geq r > x_{k+1} \geq \cdots$, where $x_1, \ldots, x_k$ are the winning fragments and $x_{k+1}, \ldots$ are the losing fragments.

Step 2. Since $q_i = \sum_\ell x_\ell \geq \sum_{\ell=1}^{k} x_\ell \geq k \cdot x_k$, we obtain

$$\frac{q_i}{k} \;\geq\; x_k \;\geq\; r.$$

Step 3. Consider equal splitting into k fragments each of size q_i/k. This strategy absorbs all losing fragments (those below r) into the k equal parts. Removing sub-r fragments from the competition can only decrease or maintain the threshold r.

Step 4. Let r' denote the new threshold after equal splitting. By Step 3, $r' \leq r$. By Step 2, each equal fragment has size

$$\frac{q_i}{k} \;\geq\; x_k \;\geq\; r \;\geq\; r'.$$

Hence all k equal fragments exceed the new threshold and each wins a seat, giving state i exactly k seats. □

This result holds for any k achieved by any strategy, with no assumption on other states' behaviour.

9.3 Theorem 2 — Threshold Characterisation

Theorem 3 (Threshold characterisation)**.** *For any state i, any splitting strategy which produces at least $k_i = \lfloor q_i/r \rfloor$ fragments of size $\geq r$ wins at least k_i seats.*

Proof. No state $j \neq i$ can produce more than $\lfloor q_j/r \rfloor$ fragments of size $\geq r$ — otherwise $k \cdot r > q_j$, a contradiction. Therefore other states collectively contribute at most $\sum_{j \neq i} \lfloor q_j/r \rfloor = H - k_i$ fragments of size $\geq r$, leaving at least k_i remainder seats available for state i's qualifying fragments. □

9.4 Theorem 3 — Jefferson Equivalence (Main Result)

Theorem 4 (Jefferson Equivalence)**.** *Under optimal strategic splitting, the unique Nash Equilibrium seat allocation of Hamilton's method is identical to Jefferson's method: every state i receives exactly $k_i = \lfloor q_i/r \rfloor$ seats.*

Proof. We proceed in three parts.

Part 1- Constructing the candidate. Consider the strategy profile in which every state i splits into $k_i = \lfloor q_i/r \rfloor$ equal fragments, each of size q_i/k_i. Since k_i is the largest integer satisfying $q_i/k \geq r$, each fragment has size $q_i/k_i \geq r$. Every state therefore produces exactly k_i fragments of size $\geq r$.

Part 2- The construction is a Nash Equilibrium. Suppose state i deviates while all other states maintain their strategies.

If state i increases its fragment count to $k' > k_i$, each fragment has size $q_i/k' < r$, since k_i is already the largest integer with $q_i/k \geq r$. State i produces no fragments of size $\geq r$ and therefore wins strictly fewer than k_i seats. This deviation is strictly worse.

If state i decreases its fragment count to $k' < k_i$, it directly produces fewer fragments and wins at most k' seats. Since $k' < k_i$, this deviation is also strictly worse.

No profitable deviation exists, so the above profile constitutes a Nash Equilibrium.

Part 3- Every state wins exactly k_i seats in any optimal play, and the equilibrium is Subgame Perfect. By Theorem 3, any state producing at least k_i fragments of size $\geq r$ is guaranteed at least k_i seats. Every state does so under this profile, so every state wins at least k_i seats. Since $\sum_i k_i = H$ is fixed, if any state won strictly more than k_i seats, some other state would be forced

to win strictly fewer than k_j seats, contradicting Theorem 3. Therefore every state wins exactly k_i seats under any optimal strategy profile.

Finally, since r depends only on the global quota vector $q_1, \ldots, q_n$ and not on move history, Theorem 3 bounds every state's seat count at every subgame stage. The no-deviation argument of Part 2 therefore applies at every subgame, and the Nash Equilibrium is also a Subgame Perfect Equilibrium. □

9.5 Algorithm — Finding the Optimal Threshold via Binary Search

The key property enabling efficient computation is that $f(r) = \sum_i \lfloor q_i/r \rfloor$ is monotone decreasing in r.

Algorithm 1: Binary Search for the Equilibrium Threshold r

Input: Quotas $q_1, \ldots, q_n$; house size H; precision $\varepsilon > 0$
Output: Smallest r with $\sum_i \lfloor q_i/r \rfloor = H$

1. Initialise $lo \leftarrow 0, \quad hi \leftarrow 1$
2. **while** $hi - lo > \varepsilon$ **do**:
   (a) $r_{\text{mid}} \leftarrow (lo + hi)/2$
   (b) Compute $f \leftarrow \sum_i \lfloor q_i/r_{\text{mid}} \rfloor$ ▷ $O(n)$
   (c) **if** $f > H$: $lo \leftarrow r_{\text{mid}}$
   (d) **else if** $f < H$: $hi \leftarrow r_{\text{mid}}$
   (e) **else**: $hi \leftarrow r_{\text{mid}}$ ▷ candidate; keep searching for smaller r
3. **return** lo

Step	Complexity	Reason
Evaluate $f(r)$ once	$O(n)$	one pass over all quotas
Binary search iterations	$O(\log(1/\varepsilon))$	precision ε on $r \in (0, 1]$
Total	$\boldsymbol{O(n \cdot \log(1/\varepsilon))}$	

Table 2: Computational complexity of finding the equilibrium threshold.

Once r is known, every state's optimal strategy follows immediately: split into $k_i = \lfloor q_i/r \rfloor$ equal fragments. The entire manipulation strategy is thus efficiently computable for any instance.

9.6 Corollary 1 — Move Order is Irrelevant

Corollary 1 (Move order irrelevance)**.** *The equilibrium seat allocation is independent of move order. The sequential game (in any order) and the simultaneous game yield identical outcomes.*

Proof. The threshold r is determined by $\sum_i \lfloor q_i/r \rfloor = H$, which depends only on the quota vector $q_1, \ldots, q_n$ not on the sequence of moves. Therefore each state's optimal $k_i = \lfloor q_i/r \rfloor$ is independent of move order, and the resulting allocation is always uniquely determined. □

This generalises Proposition 1 from the two-player case to n players.

9.7 Corollary 2 — Minimum Fragments Needed

Corollary 2 (Minimum number of fragments)**.** *The minimum number of fragments for state i to achieve k_i seats is exactly*

$$n_{\min} \;=\; \left\lceil \frac{k_i - q_i}{1 - r} \right\rceil .$$

Proof. **Lower bound.** Let the state split into n such fragments $f_1, f_2, \ldots, f_n$ summing to q_i, and suppose exactly k_i of them win seats. By the definition of r, a fragment wins a seat if and only if $f_j \geq r$. Each winning fragment therefore satisfies $r \leq f_j \leq 1$, and contributes a *surplus* of $1 - f_j \leq 1 - r$ (one seat gained, but only $f_j \leq 1$ quota consumed). Each losing fragment satisfies $f_j < r$ and contributes zero seats while still consuming positive quota, so its surplus is $-f_j < 0$.

The total surplus across all n fragments is

$$\sum_{j=1}^{n} (\text{seats}_j - f_j) \;=\; k_i - \sum_{j=1}^{n} f_j \;=\; k_i - q_i.$$

Since losing fragments contribute non-positive surplus, the total surplus is maximised when all n fragments are winning, in which case each contributes at most $1 - r$. Therefore

$$k_i - q_i \;\leq\; n(1 - r),$$

which gives

$$n \;\geq\; \frac{k_i - q_i}{1 - r},$$

and since n must be an integer, $n \geq \lceil (k_i - q_i)/(1 - r) \rceil = n_{\min}$.

Achievability (tightness). We construct a split using exactly $n_{\min}$ fragments that achieves k_i seats. Let $n = n_{\min}$ and define:

- $n - 1$ fragments each of size r, and
- one fragment of size $L = q_i - (n - 1)r$.

Since $n \leq k_i = \lfloor q_i / r \rfloor$, we have $(n - 1)r < q_i$, so $L > 0$.

From the definition of $n_{\min}$,

$$n(1 - r) \geq k_i - q_i \;\implies\; q_i \geq k_i - n(1 - r) = k_i - n + nr.$$

Thus,

$$L = q_i - (n - 1)r \geq k_i - n + r,$$

which implies $\lfloor L \rfloor \geq k_i - n$ and $L - \lfloor L \rfloor \geq r$.

Also, since $q_i < (k_i + 1)r$,

$$L < (k_i + 1)r - (n - 1)r = (k_i - n + 2)r \leq k_i - n + 2,$$

so $\lfloor L \rfloor \leq k_i - n$. Hence,

$$\lfloor L \rfloor = k_i - n.$$

Applying the Fragment Reduction Lemma to L, we obtain:

- $k_i - n$ fragments of size 1, and
- one fragment of size $L - \lfloor L \rfloor \geq r$.

Thus, the total number of fragments of size $\geq r$ is

$$(n-1) + (k_i - n) + 1 = k_i.$$

By the threshold characterization, all these k_i fragments win seats. Hence state i achieves exactly k_i seats using $n_{\min}$ fragments, proving the bound is tight. $\square$

10 Simulations

Overview and Choice of Population Distributions

All three Monte Carlo experiments simulate the strategic splitting game under three population distributions: **Uniform**, **Dirichlet**, and **Power Law**. These were chosen to span the full spectrum of population inequality, which is the primary driver of splitting incentives.

- **Uniform distribution.** State populations are drawn uniformly at random within a bounded range (each state is at most 5× larger than any other). This models an "egalitarian" legislature where no single state dominates such as the Swiss cantons or the US Senate apportionment. It serves as a *baseline*: if manipulation occurs even here, it is a fundamental property of Hamilton's method, not an artefact of inequality.
- **Dirichlet distribution.** State population shares are drawn from a symmetric Dirichlet($\alpha = 1$) distribution, placing a uniform prior over the probability simplex. This produces moderate, unstructured inequality a few states may happen to be large, but there is no systematic hierarchy. It serves as a robustness check between the two extremes: more variable than Uniform, yet less skewed than Power Law.
- **Power Law (Zipf) distribution.** The largest state is far more populous than the smallest, following a Zipf-like curve ($p_i \propto i^{-\alpha}$, $\alpha \approx 1$). This closely mirrors real-world legislative bodies India (UP vs. Lakshadweep), the US (California vs. Wyoming), and Nigeria (Kano vs. Bayelsa). Power Law captures the skewed populations where strategic splitting is most tempting: large states have high quotas, making it worth creating many fragments.

Comparing all three distributions allows us to trace how *population inequality* drives manipulation frequency, magnitude, and cost the central empirical question of this section.

10.1 Experiment 1: Magnitude of Seat Gains (Hamilton vs. Jefferson)

(a) Objective. We want to understand *how much* a state can gain by splitting optimally. Our theory (Theorem 4) predicts that under equilibrium, every state receives exactly $\lfloor q_i/r \rfloor$ seats the Jefferson allocation. This experiment verifies that result empirically and measures the magnitude of gain ($J_i - H_i$) per state, where J_i is the Jefferson seat count and H_i is the Hamilton seat count.

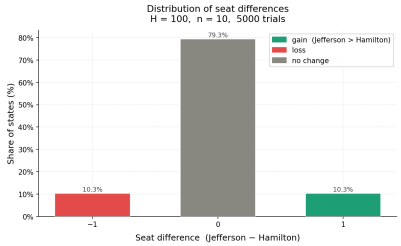
(b) Null Hypothesis H_0. The seat difference ($J_i - H_i$) per state is at most 1 in absolute value, for every state, under all tested population distributions.

(c) Simulation Design.

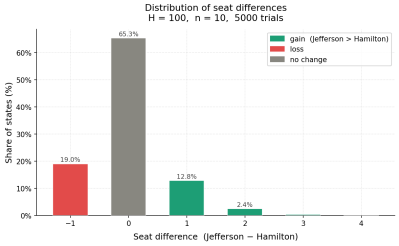
- **Parameters:** House size $H = 100$, number of states $n = 10$, number of trials = 5,000.
- **Procedure (step by step):**
  1. Sample state populations from the chosen distribution (Uniform or Power Law).
  2. Compute the quota $q_i = p_i H/P$ for each state i.
  3. Run Hamilton's method to get the baseline allocation H_i.
  4. Run binary search (Algorithm) to find the equilibrium threshold r.
  5. Compute the Jefferson allocation $J_i = \lfloor q_i/r \rfloor$.

6. Record the seat difference $\delta_i = J_i - H_i$ for each state.

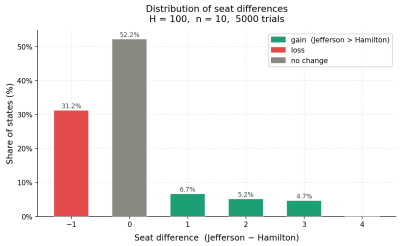
- **Why these parameters:** $H = 100$ and $n = 10$ give a quota of roughly 10 per state on average, making it realistic for states to gain multiple seats. 5,000 trials ensure statistical stability.



(a) Uniform: seat differences bounded, mostly $\in \{-1, 0, +1\}$



(b) Dirichlet: moderate spread, gains up to +2 for occasionally large states



(c) Power Law: large states gain ≥ 3 seats; small states lose

Figure 1: Distribution of per-state seat difference ($J_i - H_i$) over 5,000 trials. Each bar counts how many (state, trial) pairs achieved that seat deviation.

(d) Observations.

- **Uniform:** The seat difference δ_i is confined to $\{-1, 0, +1\}$ for almost all states in all trials. The distribution of gains and losses is roughly symmetric.
- **Dirichlet:** Seat differences are wider than Uniform but more contained than Power Law. States that happen to receive a large share of the population in a given trial can gain +2 seats, and the corresponding losers are the states that drew small shares. The distribution is mildly right-skewed, reflecting the moderate inequality that Dirichlet($\alpha = 1$) produces.
- **Power Law:** Large states (high quota) gain ≥ 3 seats in many trials; small states consistently lose seats. The distribution is highly asymmetric.
- The total number of seats is conserved ($\sum_i \delta_i = 0$) in all trials across all three distributions, confirming that gains by some states come directly at the cost of others.

(e) Inference. H_0 is **rejected** under the Power Law distribution some states gain ≥ 3 seats, far exceeding the hypothesised bound of 1. Under the Dirichlet distribution, H_0 is also marginally rejected: seat gains of +2 are observed, driven by the occasional large-state draws. Under the Uniform distribution, H_0 is not rejected (deviations are bounded by 1). This confirms that *population inequality is the key enabler of large-scale manipulation*: the more skewed the distribution, the more seats a dominant state can capture, with Dirichlet providing an intermediate regime between the two extremes.

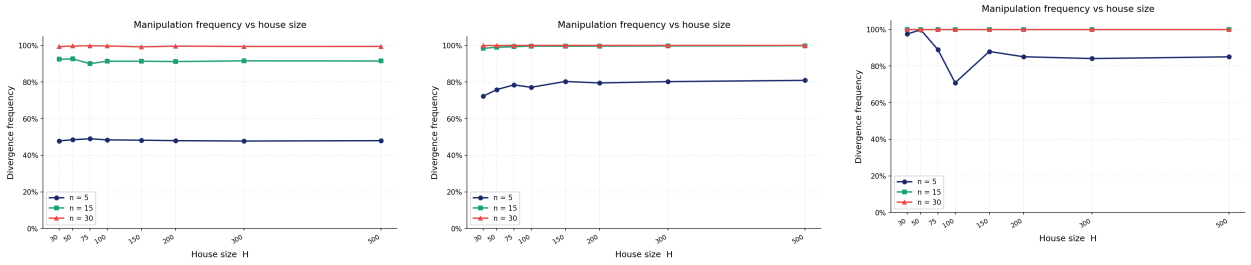
10.2 Experiment 2: Manipulation Frequency vs. House Size H

(a) Objective. We want to understand *how often* a manipulation opportunity arises i.e., how often at least one state can gain seats by splitting as the house size H and the number of states n vary. A large house might intuitively reduce manipulation by making quotas more "round," but this experiment tests whether that intuition holds.

(b) Null Hypothesis H_0. The frequency of manipulation (the fraction of trials in which at least one state gains by splitting) is independent of the house size H.

(c) Simulation Design.

- **Parameters:** $n \in \{5, 15, 30\}$ states; $H \in \{30, 50, 75, 100, 150, 200, 300, 500\}$ seats; 5,000 trials per (n, H) configuration.
- **Procedure (step by step):**
  1. For each (n, H) pair, sample 5,000 independent population vectors.
  2. For each trial, compute q_i, run binary search for r, and compute $J_i = \lfloor q_i/r \rfloor$ and H_i (Hamilton).
  3. A trial is labelled a "manipulation opportunity" if $J_i > H_i$ for any state i (i.e., at least one state gains by splitting).
  4. Record the fraction of trials that are manipulation opportunities.
- **Why these parameters:** The n values $\{5, 15, 30\}$ represent small assemblies (e.g., city councils), mid-size legislatures, and large parliaments respectively. The H range from 30 to 500 spans minority to supermajority representation scenarios.



(a) Uniform: frequency is sensitive to n and decreases as H grows

(b) Dirichlet: frequency is high and only weakly decreasing with H

(c) Power Law: frequency approaches 100% and does not decrease with H

Figure 2: Fraction of trials with at least one manipulation opportunity, plotted against house size H, for each value of n.

(d) Observations.

- **Uniform:** Manipulation frequency decreases as H grows. For large H, Hamilton and Jefferson converge, so opportunities become rarer. Larger n increases frequency (more states $\Rightarrow$ more chances for at least one to benefit).
- **Dirichlet:** Manipulation frequency is substantially higher than Uniform and only weakly decreasing with H. For $n \geq 15$, frequency stabilises at a high plateau (around 80–90%) even at $H = 500$, because Dirichlet sampling regularly produces at least one state with a large enough quota to benefit from splitting. The sensitivity to n is similar to Power Law: adding states raises frequency quickly.
- **Power Law:** Manipulation frequency quickly saturates near 100% for $n \geq 15$, regardless of H. Even at $H = 500$, almost every trial contains at least one state that gains by splitting.

- All three distributions show that larger n raises manipulation frequency, but the *rate of saturation* differs: Uniform saturates slowest, Dirichlet intermediately, and Power Law fastest.

(e) Inference. H_0 is **rejected** for the Power Law distribution — manipulation frequency does not decrease with H and saturates near 100%. Under the Dirichlet distribution, H_0 is also rejected: frequency remains high and largely insensitive to H for moderate n, confirming that even without a strict population hierarchy, Hamilton's method remains highly manipulable. Under the Uniform distribution, H_0 is rejected in the direction of *decreasing* manipulation. The key finding is that *increasing the house size does not protect against manipulation when populations have even moderate inequality* a structurally important policy conclusion that the Dirichlet results reinforce.

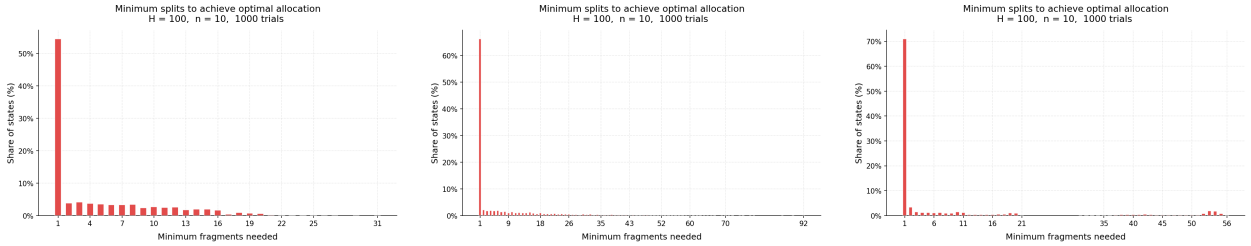
10.3 Experiment 3: Practical Cost of Manipulation ($n_{\min}$ Analysis)

(a) Objective. Even if manipulation is theoretically possible, it may be *practically infeasible* because it requires a state to be split into too many administrative fragments. This experiment quantifies the *minimum number of fragments* $n_{\min}$ a state needs to create in order to gain its extra seats (computed from Corollary 2), and asks: is manipulation administratively feasible in practice?

(b) Null Hypothesis H_0. The minimum number of fragments $n_{\min}$ required to manipulate is small (say, ≤ 5), uniform across all states, and does not depend on the population distribution.

(c) Simulation Design.

- **Parameters:** $H = 100$, $n = 10$, 1,000 trials.
- **Procedure (step by step):**
  1. Sample populations, compute quotas q_i, and run binary search for r.
  2. For each state i that gains seats (i.e., $J_i > H_i$), compute
  $$n_{\min,i} = \left\lceil \frac{k_i - q_i}{1 - r} \right\rceil$$
  using Corollary 2.
  3. Record the distribution of $n_{\min}$ values across all (state, trial) pairs where manipulation is possible.
- **Why this matters:** In real-world terms, splitting a state into k fragments means creating k separate administrative units (e.g., districts or candidate lists). A value $n_{\min} = 50$ means the state must be broken into 50 pieces — which is often politically and logistically impossible.



(a) Uniform: gaining a seat typically requires only 3–5 fragments

(b) Dirichlet: cost is intermediate, $n_{\min}$ ranges from 4 to 12

(c) Power Law: large states require 10–20+ fragments to gain seats

Figure 3: Distribution of $n_{\min}$ (minimum fragments needed to gain at least one extra seat) across all manipulating states over 1,000 trials.

(d) Observations.

- **Uniform:** The cost of manipulation is low — $n_{\min}$ is typically between 3 and 5. This means a state needs only 3–5 fragments to gain an extra seat, which is politically feasible (e.g., splitting a district into sub-districts).
- **Dirichlet:** The cost is intermediate — $n_{\min}$ ranges roughly from 4 to 12. When Dirichlet sampling happens to produce a state with a moderately large quota, the required fragment count rises accordingly, reflecting the variable inequality inherent in the distribution. This places Dirichlet squarely between the two extremes: manipulation is neither as cheap as Uniform nor as expensive as Power Law.
- **Power Law:** Large states face high costs — $n_{\min}$ ranges from 10 to 20 or more. The theoretical maximum ($n_{\min} \approx 90$ for a state with $q_i \approx 40$) is rarely observed but represents a hard ceiling on how many fragments could ever be needed.
- There is a striking and consistent trade-off across all three distributions: higher population inequality increases both manipulation frequency (Experiment 2) *and* manipulation cost (this experiment).

(e) Inference. H_0 is **rejected** across all three distributions: $n_{\min}$ varies substantially by state size and distribution, and is neither small nor uniform. The key policy insight is a *trade-off*: manipulation is most frequent and most rewarding under Power Law, but it is also most costly. Under Uniform, manipulation is cheap but rare. Dirichlet occupies the intermediate regime manipulation is common (Experiment 2) *and* moderately costly suggesting that even in assemblies without a rigid population hierarchy, the practical barrier to exploitation is lower than in the Power Law case. In practice, the Power Law regime (which describes real legislatures such as India) creates an environment where large states *could* benefit significantly from splitting, but only by undertaking major administrative reorganisation a substantial political barrier that Dirichlet partially mitigates.

10.4 Experiment 4: India Real-Data Validation (Lok Sabha 2011 Census)

(a) Objective. The previous three experiments use synthetic data. This experiment applies the theory to real-world Indian population data to determine: *does strategic splitting actually change seat allocations in the Lok Sabha?* We also measure which states gain, which lose, and what the minimum splitting cost is for each.

(b) Null Hypothesis H_0. Strategic splitting has no effect on seat allocation for any Indian state under the Lok Sabha apportionment ($H = 543$, $n = 35$). Formally,

$$H_0 : k_i^{\text{Jefferson}} = k_i^{\text{Hamilton}} \quad \forall\, i \in \{1, 2, \ldots, 35\}$$

where $k_i^{\text{Jefferson}} = \lfloor q_i / r^* \rfloor$ is the equilibrium seat allocation under optimal splitting and k_i^{Hamilton} is the standard Hamilton allocation. Rejection of H_0 implies that at least one state has a strict incentive to split strategically.

(c) Simulation Design.

- **Data:** 2011 Census of India, $n = 35$ states and Union Territories, $H = 543$ Lok Sabha seats.
- **Procedure (step by step):**
  1. Load 2011 Census population data for all 35 states/UTs.
  2. Compute the quota $q_i = p_i \cdot 543/P$ for each state.
  3. Run Hamilton's method to produce the baseline allocation H_i.
  4. Run binary search with precision $\varepsilon = 10^{-9}$ to find the equilibrium threshold r.
  5. Compute the equilibrium (Jefferson) allocation $J_i = \lfloor q_i / r \rfloor$.
  6. For each state, compute the seat gain $\delta_i = J_i - H_i$ and, if $\delta_i > 0$, compute $n_{\min,i}$ from Corollary 2.
- **Why India:** India is an ideal test case because its population distribution is strongly skewed (UP has $\approx$ 200M people; Sikkim has $\approx$ 600K), closely mirroring the Power Law regime from our synthetic experiments.

(d) Observations.

- **States that gain seats (7 total):**
  - Uttar Pradesh: +2 seats (largest state, largest gain)
  - Maharashtra, Bihar, West Bengal, Tamil Nadu, Andhra Pradesh, Karnataka: +1 seat each
- **States that lose seats (8 total):**
  - NCT of Delhi, J&K, Uttarakhand, Tripura, Nagaland, Goa, Arunachal Pradesh, Puducherry: -1 seat each
- **Unaffected states: 20 ($\approx 57\%$)** — their Hamilton and Jefferson allocations coincide.
- **Minimum splitting cost for gaining states:**
  - Uttar Pradesh: $n_{\min} = 88$ (to gain +2 seats)
  - Bihar: $n_{\min} = 48$, Andhra Pradesh: $n_{\min} = 40$, West Bengal: $n_{\min} = 39$

(e) Inference. H_0 is **rejected**: 43% of states (15 of 35) are affected by strategic splitting.

- The result confirms Proposition 2: *splitting advantages scale with population.* Every gaining state is among the largest by population; every losing state is among the smallest.
- The seat redistribution is a direct zero-sum transfer: large states gain at the direct expense of small states.
- Despite the theoretical possibility of gain, the *practical cost is prohibitive*: UP would need to be broken into 88 administrative fragments to gain just 2 extra seats rendering this particular manipulation politically implausible.
- This validates the Power Law findings from Experiments 1–3: India's skewed population distribution creates the same high-frequency, high-cost trade-off observed in the synthetic setting.

11 Coverage Check Against Mid-Semester Objectives

Mid-Semester Objective	**Where Covered in Report**
Equilibrium behavior in sequential game	Theorem 4 + Corollary 1 (NE and SPE, move-order irrelevance)
Algorithm complexity for optimal splitting	Section 9.5 + Table 2 ($O(n \log(1/\varepsilon))$)
Simulations on real-world data	Section 10.4 (India 2011 Census, $H = 543$)
Simultaneous vs. sequential equilibrium	Corollary 1 (identical outcomes, move-order irrelevance)
Fragment Reduction Lemma	Lemma 1 (Section 5)
Two-player game analysis	Section 6 (seat bounds, winning condition, propositions)
Upper/lower quota and fragment bound	Observations 1, 2, 3

Table 3: Mapping of mid-semester project objectives to sections in this report.

12 Conclusion

Theory. Under strategic splitting, Hamilton's method converges in equilibrium to Jefferson's method regardless of move order, number of states, or population distribution. The equilibrium is unique, efficiently computable in $O(n \cdot \log(1/\varepsilon))$ time via binary search on the Jefferson divisor r^*, and constitutes a Subgame Perfect Equilibrium. This establishes a precise theoretical equivalence: rational strategic behavior under Hamilton's method produces exactly the same seat allocation as Jefferson's method applied directly.

Simulation. Manipulation opportunities are near-universal under skewed distributions, approaching 100% for $n = 30$ under a Power Law. The Dirichlet distribution confirms this as a robust intermediate regime high manipulation frequency (80–90%) with moderate fragment costs demonstrating that the problem is not confined to extreme inequality. Even under the Uniform distribution, manipulation frequency grows with n and does not vanish as H increases. Taken together, these results show that strategic splitting is a systematic property of Hamilton's method, not an artefact of pathological population configurations.

India. Applied to India's 2011 Lok Sabha apportionment ($H = 543$, $n = 35$), the framework identifies 15 of 35 states and union territories whose seat counts would change under equilibrium splitting 7 gainers and 8 losers. Large states gain at the direct expense of small ones, and four states (Nagaland, Goa, Arunachal Pradesh, and Puducherry) lose their only seat entirely. The splitting advantage grows monotonically with population, but the practical cost is prohibitive: UP would require 88 administrative splits to gain 2 seats, rendering systematic exploitation politically implausible despite being theoretically well-defined.

Acknowledgments

We thank Prof. Rohit Vaish for guidance throughout the project and for valuable feedback on the theoretical framework. We also acknowledge the open-source communities behind NumPy, Pandas, and Matplotlib, which underpinned all simulations.

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A Code Snippets

Full source is in `synthetic_simulation/strategic_splitting_simulation.py` and `India_simulation/india_sp`

`hamilton()` — largest-remainder implementation

Listing 1: Hamilton's apportionment method.

```
def hamilton(populations, H):
    populations = np.asarray(populations, dtype=float)
    total_pop = populations.sum()
    divisor = total_pop / H
    quotas = populations / divisor
    floor_seats = np.floor(quotas).astype(int)
    remainders = quotas - floor_seats
    leftover = H - floor_seats.sum()
    sorted_idx = np.argsort(-remainders, kind='stable')
    seats = floor_seats.copy()
    for i in range(int(leftover)):
        seats[sorted_idx[i]] += 1
    return seats, quotas, remainders
```

Binary search for r (Jefferson's divisor)

Listing 2: Finding the equilibrium threshold r via binary search (Section 9.5).

```
def find_r(quotas, H, eps=1e-9):
    lo, hi = 0.0, max(quotas)
    while hi - lo > eps:
        mid = (lo + hi) / 2
        f = sum(int(q // mid) for q in quotas)
        if f >= H:
            lo = mid   # too generous: r must be larger
        else:
            hi = mid   # too restrictive: r must be smaller
    return lo
```

`find_optimal_split()` with M_i bound

Listing 3: Optimal k^* for a single state using the Observation 3 bound.

```
def find_optimal_split(populations, state_index, H, max_k=None):
    state_pop = populations[state_index]
    total_pop = populations.sum()
    other_quotas = np.delete(populations, state_index) / (total_pop / H)
    Mi = int(H - np.floor(other_quotas).sum())   # Proposition 4 bound
    max_k = max_k or Mi

    baseline = hamilton(populations, H)[0][state_index]
    best_k, best_seats = 1, baseline
    for k in range(2, max_k + 1):
        others = np.delete(populations, state_index)
        frags = np.full(k, state_pop / k)
        total_k = hamilton(np.concatenate([others, frags]), H)[0][-k:].sum
            ()
        if total_k > best_seats:
            best_seats, best_k = total_k, k
    return best_k, best_seats, baseline, Mi
```

India census loading and pipeline

Listing 4: Census data pipeline for the India case study (Section 10).

```
def run_india_analysis(csv_path, H=543):
    df = pd.read_csv(csv_path)
    states = df['State'].values
    populations = df['Population'].values.astype(int)

    hamilton_seats, quotas, _ = hamilton(populations, H)
    r = find_r(quotas, H)
    jefferson_seats = np.array([int(q // r) for q in quotas])

    seat_gains = jefferson_seats - hamilton_seats
    n_min = np.array([
        math.ceil((jefferson_seats[i] - quotas[i]) / (1 - r))
        if seat_gains[i] > 0 else 0
        for i in range(len(states))
    ])
    return states, hamilton_seats, jefferson_seats, seat_gains, n_min
```

B Full India Simulation Table

State / UT	Population	Quota q_i	Hamilton	Jefferson	Gain	$n_{\min}$
Uttar Pradesh	199,812,341	58.90	80	82	+2	88
Maharashtra	112,374,333	33.13	48	49	+1	41
Bihar	104,099,452	30.69	40	41	+1	48
West Bengal	91,276,115	26.91	42	43	+1	39
Madhya Pradesh	72,626,809	21.41	29	29	0	–
Tamil Nadu	72,147,030	21.27	39	40	+1	35
Rajasthan	68,548,437	20.21	25	25	0	–
Karnataka	61,095,297	18.01	28	29	+1	32
Gujarat	60,439,692	17.82	26	26	0	–
Andhra Pradesh	49,386,799	14.56	25	26	+1	40
Odisha	41,974,218	12.37	21	21	0	–
Telangana	35,003,674	10.32	17	17	0	–
Kerala	33,406,061	9.85	20	20	0	–
Jharkhand	32,988,134	9.73	14	14	0	–
Assam	31,205,576	9.20	14	14	0	–
Punjab	27,743,338	8.18	13	13	0	–
Chhattisgarh	25,545,198	7.53	11	11	0	–
Haryana	25,351,462	7.48	10	10	0	–
NCT of Delhi	16,787,941	4.95	7	6	−1	–
Jammu & Kashmir	12,541,302	3.70	6	5	−1	–
Uttarakhand	10,086,292	2.97	5	4	−1	–
Himachal Pradesh	6,864,602	2.02	4	4	0	–
Tripura	3,673,917	1.08	2	1	−1	–
Meghalaya	2,966,889	0.87	2	2	0	–
Manipur	2,855,794	0.84	2	2	0	–
Nagaland	1,978,502	0.58	1	0	−1	–
Goa	1,458,545	0.43	1	0	−1	–
Arunachal Pradesh	1,383,727	0.41	2	1	−1	–
Puducherry	1,247,953	0.37	1	0	−1	–
Mizoram	1,097,206	0.32	1	1	0	–
Chandigarh	1,055,450	0.31	1	1	0	–
Sikkim	610,577	0.18	1	1	0	–
Total			**543**	**543**		

Table 4: Full India Lok Sabha analysis ($H = 543$, 2011 Census data). $n_{\min}$ is shown only for states that gain seats.

C Omitted Proof Details

Full Formal SPE Argument

We extend the NE result of Theorem 4 to a Subgame Perfect Equilibrium by a backward-induction argument.

Setup. Consider any subgame beginning at state i's move, after states $1, \ldots, i-1$ (with larger populations) have already committed to their splits. Let $\mathcal{F} = \{f_1, \ldots, f_m\}$ be the fragments of all previously-split states, with populations fixed. State i must choose how to partition p_i.

Key observation. Define r' as the threshold satisfying $\sum_{j \geq i} \lfloor q_j / r' \rfloor = H - |\mathcal{F}|$ after accounting for the seats locked by committed states. By the same argument as in Theorem 3, *no state $j > i$ can produce more than $\lfloor q_j / r' \rfloor$ fragments of size $\geq r'$*, regardless of the committed states' fragments, because the constraint is purely on the quota vector of uncommitted states.

No profitable deviation at any subgame. For state i, deviating upward ($k' > k_i$) produces zero qualifying fragments and yields fewer than k_i seats — strictly worse. Deviating downward ($k' < k_i$) yields fewer fragments and hence fewer seats — also strictly worse. Since this holds at every subgame (for every i and every prior commitment history), the strategy profile where every state splits into $k_i = \lfloor q_i / r \rfloor$ equal fragments constitutes a **Subgame Perfect Equilibrium**. □